Classification

Big Data y Machine Learning para Economía Aplicada

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Agenda

1. Previously...
2. Classification
3. Risk, Probability, and Classification
4. Probability Estimation
   - LPM
   - Logit
   - Probit
   - Logit EN

El objetivo

- Para cualquier variable aleatoria continua y y vector de covariables W, tenemos:

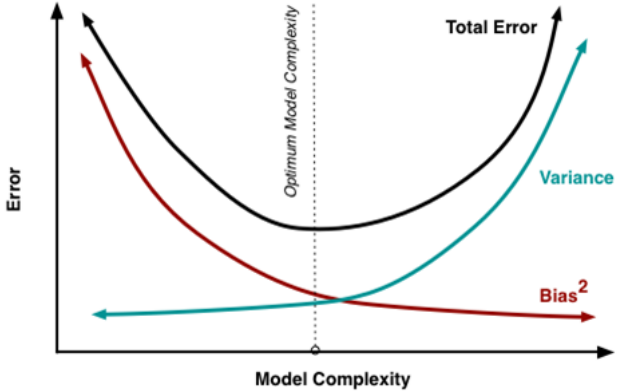
$$y = m(W) + u$$

- Bajo pérdida cuadrática, para cualquier función medible $m(W)$:

$$\hat{m} = \underset{m}{\operatorname{argmin}} \left\{ E\left[L(y, m(W)) \right] \right\}$$

Error de predicción fuera de muestra

- ML nos interesa la predicción fuera de muestra



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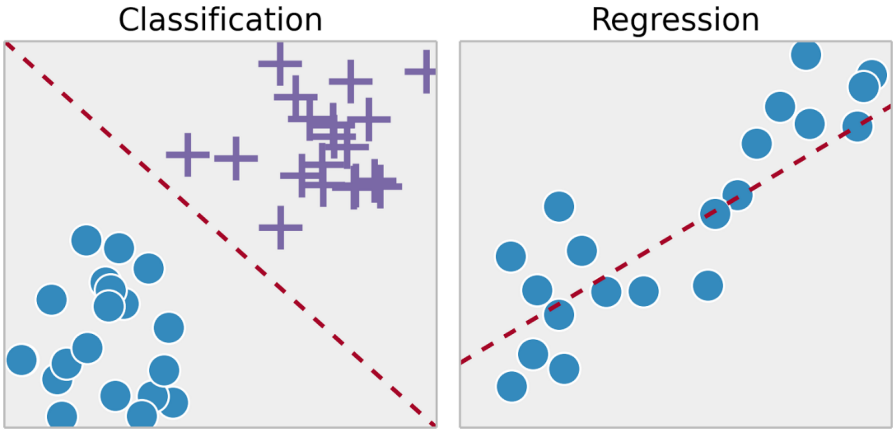
Classification



Classification: Motivation

- Many predictive questions are about classification
  - Accept someone to a graduate program or no
  - A household is bellow the poverty line
  - An individual is unemployed

Classification: Motivation

- y represents membership in a category: $y \in \{1, 2, ..., n\}$
  - Qualitative (e.g., unemployed, employed, etc)
  - Not necessarily ordered

The prediction question is, given a new X,
what is our best guess at the response category $\hat{y}$

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Risk, Probability, and Classification

- Two states of nature $Y \rightarrow i \in \{0, 1\}$
- Two actions $(\hat{Y}) \rightarrow j \in \{0, 1\}$

		$\hat{Y}$	
		0	1
Y	0	True Negative	False Positive
	1	False Negative	True Positive

Risk, Probability, and Classification

- Two actions $\hat{Y} \rightarrow j \in \{0, 1\}$
- Two states of nature $Y \rightarrow i \in \{0, 1\}$
- Probabilities
  - $p = Pr(Y = 1|X)$
  - $1 - p = Pr(Y = 0|X)$

Risk, Probability, and Classification

- Actions have costs associated to them
- Loss: $L(i,j)$, penalizes being in bin i,j
  - We define $L(i,j)$

$$L(i,j) = \begin{cases} 1 & i \neq j \\ 0 & i = j \end{cases} \tag{1}$$

- Risk: expected loss of taking action j

Bayes classifier

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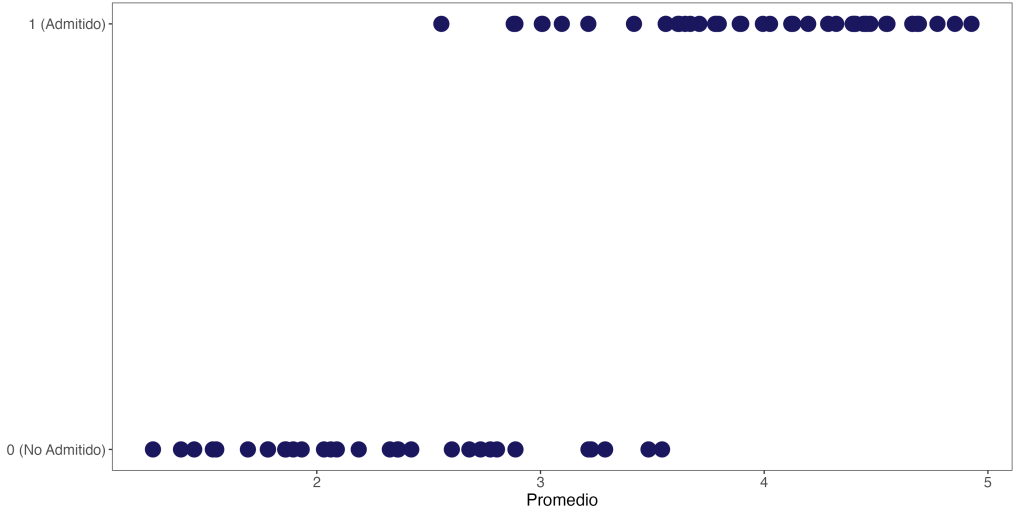
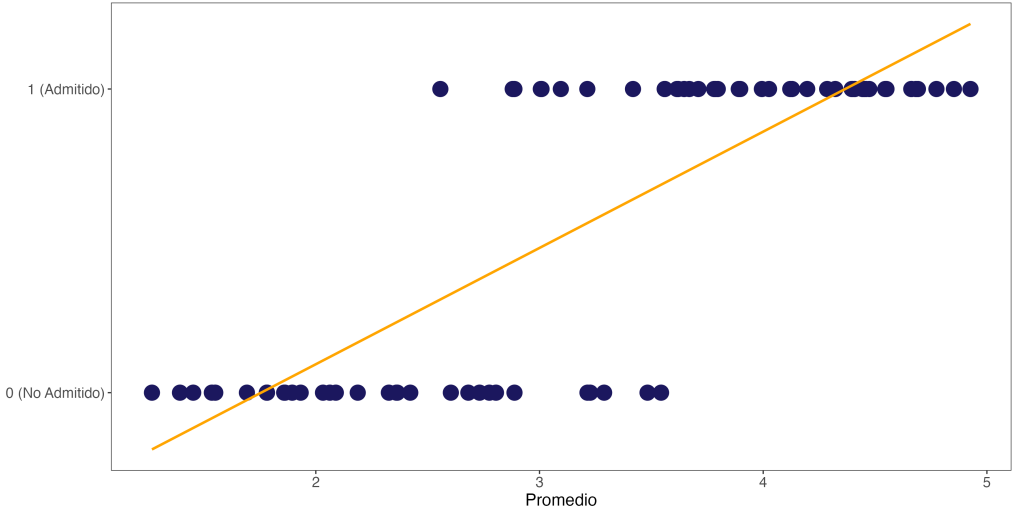
Setup

- Y is a binary random variable$\{0, 1\}$
- X is a vector of K predictors
- $p = Pr(Y = 1|X)$

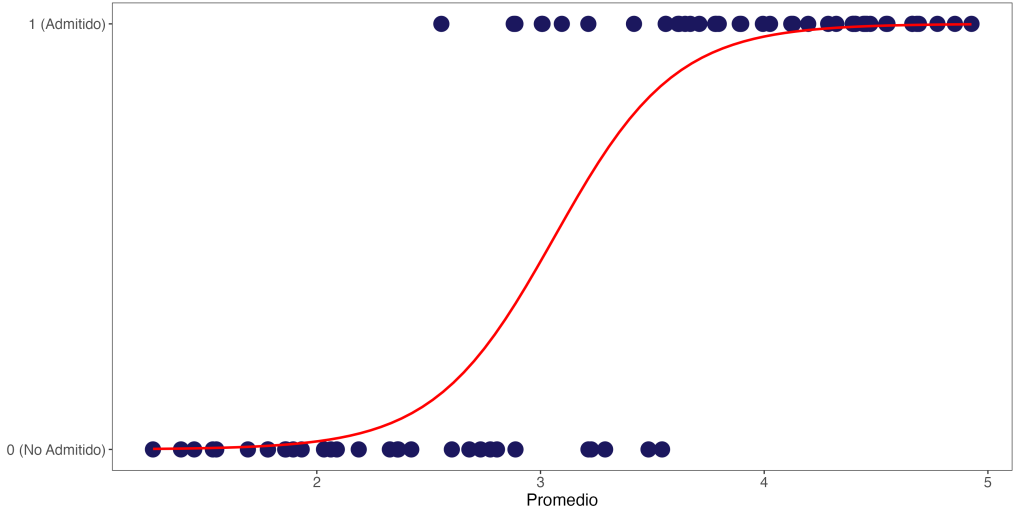
$Pr(Y = 1|X)$



$Pr(Y = 1|X)$



$$Pr(Y = 1|X)$$



$Pr(Y = 1|X)$

Logit

- We have a sample of observations $(y_i, X_i); i = 1, \ldots, n$, where $y_i \in \{0, 1\}$ and we assume are iid

$Pr(Y = 1|X)$

Logit

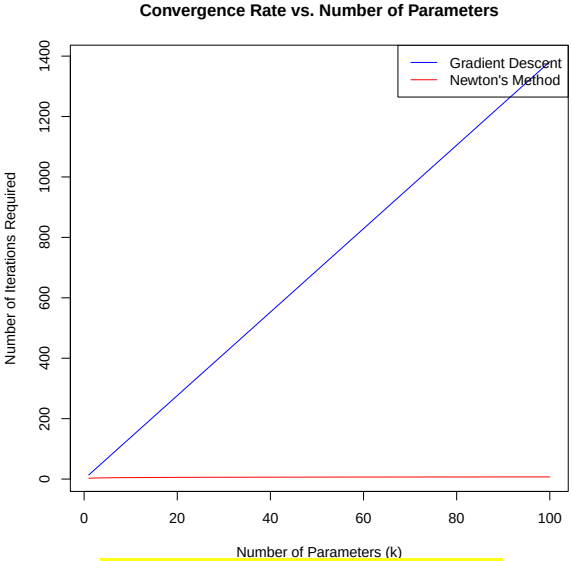
$Pr(Y = 1|X)$

Logit

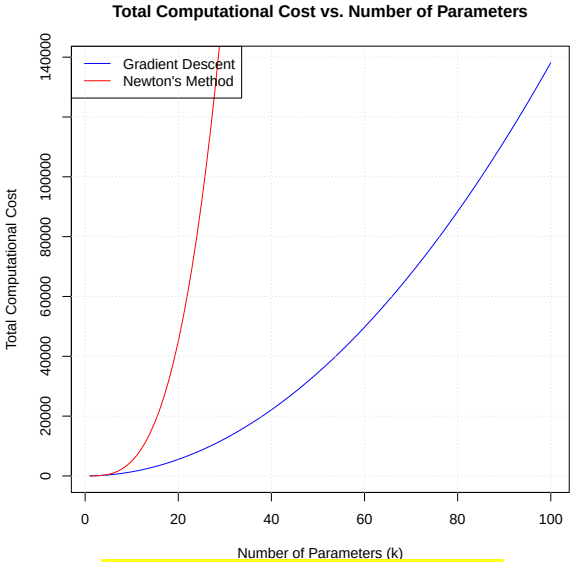
Gradient Descent vs Newton's Method

Gradient Descent vs Newton's Method

Gradient Descent vs Newton's Method



Gradient Descent vs Newton's Method



Probit

- $Pr(y = 1|X) = \Phi(X'\beta)$ where Φ is the standard normal cdf.

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Probit

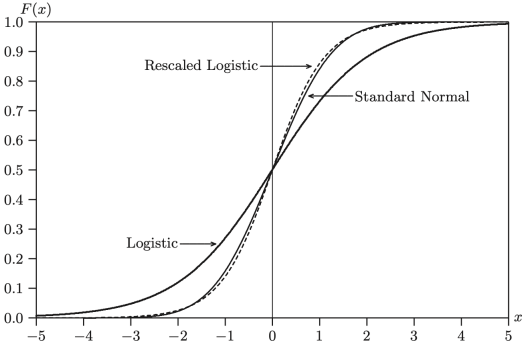

Figure 11.1 Alternative choices for $F(x)$

Probit

- $Pr(y = 1|X) = \Phi(X'\beta)$ where Φ is the standard normal cdf.
- In practice, the probit and logit models generally yield very similar predicted probabilities,
- There are practical reasons for favoring one or the other in some cases for mathematical convenience, in other computational convenience, but it is difficult to justify the choice of one distribution or another on theoretical grounds.

Logit EN

Summary

- We observe (y_i, X_i) $i = 1, \ldots, n$
- Generate probabilities
  - Logit (example)

$$p_i = \frac{e^{X_i\beta}}{1 + e^{X_i\beta}} \tag{2}$$

  - Predict

$$\hat{p}_i = \frac{e^{X_i\hat{\beta}}}{1 + e^{X_i\hat{\beta}}} \tag{3}$$

- Classification

$$\hat{y}_i = 1[\hat{p}_i > 0.5] \tag{4}$$

Accuracy

		$\hat{y}_i$	
		0	1
y_i	0	TN	FP
	1	FN	TP

- Accuracy: the fraction of predictions our model got right.

$$\frac{TP + TN}{TP + TN + FN + FP} \tag{5}$$

Example

photo from https://www.dailydot.com/parsec/batman-1966-labels-tumblr-twitter-vine/